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Introduction to Topology: Exercises

September 23, 2026

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1

Topological Spaces and Continuous Functions

1.1 Warm-ups

Q1. Consider the Sierpiński space with $X = \{0, 1\}$ and $\mathcal{O} = \{\emptyset, \{0\}, \{0, 1\}\}$.

1. Calculate the closure of $\{1\}$.
2. Is $(X, \mathcal{O})$ Hausdorff?
3. Consider $\mathcal{O}' = \{\emptyset, \{1\}, \{0, 1\}\}$. Show that $\mathcal{O}$ and $\mathcal{O}'$ are homeomorphic by exhibiting a continuous function $f : (X, \mathcal{O}) \rightarrow (X, \mathcal{O}')$.
4. Show that all sequences in $(X, \mathcal{O})$ converge to 1.
5. Describe the set of sequences in $(X, \mathcal{O})$ that converge to 0.
6. Exhibit a sequence that converges both to 0 and 1 in $(X, \mathcal{O})$.
7. Show that there are only three continuous functions from $(X, \mathcal{O})$ to itself.

Q2. Let $f : X \rightarrow Y$ be a function and suppose that Y is endowed with a topology $\mathcal{O}$. Show that if $f^{-1}(\mathcal{O}) = \{f^{-1}(O) \subset X ; O \in \mathcal{O}\}$, then $f^{-1}(\mathcal{O})$ is a topology on X . Show that this is the coarsest topology on X that makes the function f continuous.

The above topology is called the initial topology on X induced by f .

Q3. Show that given a family of topologies $\{(X, \mathcal{O}_\alpha)\}_{\alpha \in \mathcal{A}}$, their intersection, in which $A \subset X$ is open if and only if it is open in all $\mathcal{O}_\alpha$, is still a topology. Argue that replacing the intersection by the union would not work.

Q4. Show that the following are equivalent definitions for “ $\mathcal{O}_1$ is finer than $\mathcal{O}_2$ ”:

1. Every closed set of $\mathcal{O}_2$ is closed in $\mathcal{O}_1$;
2. The identity map $(E, \mathcal{O}_1) \rightarrow (E, \mathcal{O}_2)$ is continuous;
3. For any $x \in X$, a neighborhood of x in $\mathcal{O}_2$ is a neighborhood of x in $\mathcal{O}_1$.

Q5. Suppose that $(X, \mathcal{O}_1)$ is finer than $(X, \mathcal{O}_2)$. Show that:

1. If $\mathcal{O}_2$ is Hausdorff, then $\mathcal{O}_1$ is also Hausdorff;
2. If $\mathcal{O}_1$ is separable (i.e. has a countable dense subset), then $\mathcal{O}_2$ is also separable;
3. For all $A \subset X$, its closure in $\mathcal{O}_1$ is contained in its closure in $\mathcal{O}_2$, and its interior in $\mathcal{O}_2$ is included in its interior in $\mathcal{O}_1$.

Q6. Show that X is Hausdorff if and only if the diagonal $\Delta = \{(x, x) \mid x \in X\} \subset X \times X$ is closed in the product topology of $X \times X$.

1.2 Main exercise: subspace topology

Let $(X, \mathcal{O}_X)$ be a topological space and $A \subset X$. Denote by $\iota : A \rightarrow X$ the inclusion map. We define the subspace topology of A induced by the topology of X by declaring that $U \subset A$ is open if and only if there exists some open set O of X such that $U = O \cap A$.

Q1. Show that $F \subset A$ is closed in A if and only if there exists some closed set F' of X such that $F = F' \cap A$.

Q2. Show that if $\mathcal{B}$ is a basis of X , then $\{O \cap A \mid O \in \mathcal{B}\}$ is a basis of A .

Q3. Show that if A is open (resp. closed) in X if and only if all open (resp. closed) sets U of A is open (resp. closed) in X .

Q4. Show that the inclusion map $\iota : A \rightarrow X$ is continuous and that the subspace topology is the same as the weak/initial topology induced by ι (see an exercise before for the weak topology).

Q5. If we have a sequence of subspaces $A \subset B \subset X$ (B is also equipped with the subspace topology induced by X), then the closure of A in B is the intersection of the closure of A in X with B .

Q6. Show that if $\bar{A}$ is the closure of A in X , then A is dense in $\bar{A}$ (with the induced topology of $\bar{A}$ on A).

Q7. Show that if X is a metric space with distance d , then the subspace topology on A is also a metric topological space with the restriction of d to A .

Q8. If X is a Hausdorff metric space, then A with the induced topology is also Hausdorff.

***Q9.** Give an example where the subspace $A \subset X$ is Hausdorff but two points in A do not have disjoint neighborhoods in X . You can consider $Y = X \cup \{\omega\}$ where ω is an extra point, and any non-empty open set in Y is of the form $O \cup \{\omega\}$ for O open in X .

1.3 A little more: metric space exercises

Q1. Show the following theorem.

Every open subset U of $\mathbb{R}$ can be uniquely expressed as a countable union of disjoint open intervals. The end points of the intervals do not belong to U .

Hint: you can consider the maximal open interval $I_x \subset U$ containing a point $x \in U$.

Q2. Recall that a distance d on a set X is a function $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ such that¹

1. Diagonale condition: $d(x, x) = 0$ for all $x \in X$;
2. Separation condition: if $d(x, y) = 0$ then $x = y$;
3. Symmetry condition: $d(x, y) = d(y, x)$ for all $x, y \in X$;
4. Triangular inequality: $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in X$.

We call (X, d) a metric space. An ultra-metric is a distance where we replace condition 4 by the stronger condition that

$$\forall x, y, z \in X, \quad d(x, z) \leq \max(d(x, y), d(y, z)).$$

For example the usual p -adic metric is ultrametric.

Let d be an ultrametric and show the following:

¹ A pseudo-distance is such that conditions 1, 2, 4 are verified: we will not use this in this exercise.

1. If $d(x, y) \neq d(x, z)$ then $d(y, z) = \max(d(x, y), d(x, z))$.
2. Every point in an open ball is the center of this ball.
3. Every point in an open ball is the center of the sphere defined by this ball.
4. For any two open balls, they are either disjoint, or one is contained in the other.

Q3. Suppose that (X, d) is a metric space. Show that $d^*(x, y) = \min(d(x, y), 1)$ and $d^{**}(x, y) = \frac{d(x, y)}{1+d(x, y)}$ are both metrics and they induce the same topology as d .

Note that the notion of boundedness depends on the choice of the metric, not on the topology.

1.4 Colles

C1. Usual metric topology of $\mathbb{R}^n$.

Recall what is the usual metric topology of $\mathbb{R}^n$. Show that any union of open sets in $\mathbb{R}^n$ is open. Show that intersections of open sets in $\mathbb{R}^n$ might not be open.

C2. Reciprocal image and union/intersection.

Recall what is the reciprocal image of a function. Show that this operation commutes with the operations of union and intersection. What about direct image?

C3. Subspace of a vector space generated by a family of vectors.

Recall what is the vector space generated by a family of vectors. Give an abstract construction using the stability of vector spaces under intersection, and a concrete construction using finite linear combination.

C4. Closed sets in $\mathbb{R}$.

Show that a set F is closed in $\mathbb{R}$ under the usual metric topology if and only if all convergent sequences of F have their limit in F .

C5. Intersection of topologies.

Recall that the intersection of a family of topologies on the same space X is still a topology. Recall what is the topology generated by a family of subsets of X . Is there a concrete construction of this topology using finite intersections?

C6. Initial topology.

Show that if $f : X \rightarrow (Y, \mathcal{O})$ is a map into a topological space Y , then the family of sets $f^{-1}(O)$ with $O \in \mathcal{O}$ defines a topology on X . Show that it is the most coarse topology on X making f continuous: this is called the initial topology. Study the case where f is an inclusion map, for example $f : \mathbb{R} \rightarrow \mathbb{R}^2$ with the Euclidean topology on $\mathbb{R}^2$. Can you do the other way and pushforward a topology on X to define the “final topology” on Y ?

C7. Cocountable topology.

Let X be a set. Show that the following is a topology: all non-trivial open sets are complement of some finite subset of X . This is called the cofinite topology on X . Show that the sequence $\{1, 2, 3, 4, \dots\}$ in the cofinite topology on $\mathbb{R}$ converges to every point in $\mathbb{R}$. Can you define the cocountable topology on X ? Now take $X = \mathbb{R}$ equipped with the cocountable topology, and consider $(-1, 1) \subset \mathbb{R}$. Show that $\overline{B} = \mathbb{R}$. Show that no sequence in $(-1, 1)$ converges to $2 \in \mathbb{R}$.

C8. Zariski topology.

Let $\mathbb{K}$ be a field, $n \in \mathbb{N}$ a non-negative integer, and $\mathbb{A}_n(\mathbb{K}) = \mathbb{K}^n$. A Zariski-closed set of $\mathbb{A}_n(\mathbb{K})$ is such that

$$F = \{x \in \mathbb{A}_n(\mathbb{K}) ; \forall i \in I, P_i(x) = 0\},$$

where $(P_i)_{i \in I}$ is a family of polynomials on $\mathbb{A}_n(\mathbb{K})$. Show that this indeed defines a topology (called of course the Zariski topology).

Show that Zariski-open sets in $\mathbb{R}^2$ are dense in $\mathbb{R}^2$. Find the Zariski-closure of $\{(0, \frac{1}{n}) ; n \geq 0\} \subset \mathbb{R}^2$.

There is a famous theorem of David Hilbert (known as Nullstellensatz) shows that any Zariski-closed set of $\mathbb{A}_n(\mathbb{K})$ is the common zero set of a *finite* family of polynomials.

C9. Basis of topology.

Recall the definition of a basis of topology. Recall also the characterization thereof. What happens if any conditions is removed? Show that equivalence of the definition and the characterization.

2

Connectedness

2.1 Warm-ups

Q1. Show that $\mathbb{R}$ and $\mathbb{R}^2$ have the same cardinal, but are not homeomorphic. You can consider the image of $\mathbb{R}^2 \setminus \{0\}$.

Q2. Prove that there is no continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(\mathbb{P}) \subset \mathbb{Q}$ and $f(\mathbb{Q}) \subset \mathbb{P}$ (where $\mathbb{P}$ is the set of prime integers). You can consider $f(\mathbb{R})$. Alternatively, consider the restriction to $[0, 1]$ of $\frac{1}{1+|f|}$.

Q3. Show that a connected space that is locally path connected is path connected. You can consider the “path-connected” component of a point.

2.2 Main exercise: arithmetic progression topologies

We study three arithmetic progression topologies on the set $\mathbb{Z}$ or the set $\mathbb{Z}_{>0}$. They are all obtained by taking as a basis a suitable collection of arithmetic progressions.

Part A: the Furstenberg topology on $\mathbb{Z}$

We define a topology on $\mathbb{Z}$ by using the (non-trivial) arithmetic progressions $S(a, b) = a\mathbb{Z} + b = \{an + b \mid n \in \mathbb{Z}\}$ with $a \neq 0$ as a basis. That is, a set on $\mathbb{Z}$ is open if and only if it is $\emptyset, \mathbb{Z}$ or can be written as a union of arithmetic sequences $S(a, b)$.

A1. Show that $U \subset \mathbb{Z}$ is open if and only if for any $x \in U$, there is some non-zero $a \in \mathbb{Z}$ such that $S(a, x) \subset U$.

A2. Show that this defines a topology on $\mathbb{Z}$, i.e. verify the axioms for a topology.

This topology is known as the Furstenberg topology on $\mathbb{Z}$ or the evenly spaced integer topology.

A3. Show that a non-empty finite set in $\mathbb{Z}$ cannot be open. In particular, $\mathbb{Z} \setminus \{-1, 1\}$ is not closed.

A4. Show that $S(a, b)$ is both open and closed (for $a \neq 0$).

A5. Show that the union over prime numbers $\bigcup_{p \in \mathbb{P}} S(p, 0)$ is $\mathbb{Z} \setminus \{-1, 1\}$.

A6. Show that there are infinitely many prime numbers.

This is Furstenberg's topological proof on the infiniteness of prime numbers.

Part B: the Golomb topology and the Kirch topology on $\mathbb{Z}_{>0}$

We define a topology on $\mathbb{Z}_{>0}$ by using the (non-trivial) arithmetic progressions $a\mathbb{N} + b = \{an + b \mid n \in \mathbb{N}\}$ with a, b relatively prime as a basis. Here the convention is that $\mathbb{N} = \{0, 1, 2, \dots\} \neq \mathbb{Z}_{>0}$.

B1. Show that we can restrict the basis to the pairs with $b < a$ and b, a relatively prime.

B2. Show that this defines indeed a topology on $\mathbb{Z}_{>0}$.

This topology is known as the Golomb topology on $\mathbb{Z}_{>0}$ or the relative prime integer topology.

We define another topology on $\mathbb{Z}_{>0}$ by using the (non-trivial) arithmetic progressions $p\mathbb{N} + b = \{pn + b \mid n \in \mathbb{N}\}$ with p prime and p not dividing b as a (pre)basis. Here the convention is that $\mathbb{N} = \{0, 1, 2, \dots\} \neq \mathbb{Z}_{>0}$.

B3. Show that we can restrict the basis to the pairs with $b < p$ with the same conditions.

B4. Show that this defines indeed a topology on $\mathbb{Z}_{>0}$.

This topology is known as the Kirch topology on $\mathbb{Z}_{>0}$ or the prime integer topology.

Part C: topological properties of arithmetic progression topologies

This part explores the different topological properties of the above defined spaces.

C1. Show that the Furstenberg topology is Hausdorff and regular. It is metrizable, but not completely metrizable. Indeed, it is homeomorphic to the rational numbers $\mathbb{Q}$ with the subspace topology inherited from $\mathbb{R}$.

C2. Show that the Golomb topology and the Kirch topology are Hausdorff, but not regular.

C3. Show that the Furstenberg topology is totally disconnected, the Golomb topology is connected, but not locally connected and the Kirch topology is both connected and locally connected.

2.3 A little more: metric space exercises

Q1. Show that a connected metric space with more than one point is uncountable.

Q2. Show that the general linear group $GL(n, \mathbb{R})$ (that is, the group of $n \times n$ real, invertible matrices) consists of two connected components: the one with matrices of positive determinant and the other of negative determinant. In particular, it is not connected. Show that in contrast, $GL(n, \mathbb{C})$ is connected.

Q3. Show that the topologist's sine curve is connected but not path-connected.

2.4 Colles

C1. Separable space.

Recall that a topological space X is called separable if it has a countable dense subset. Show that a topological space X with a countable basis is separable.

Now consider $\mathbb{R}$ with basis $\mathcal{B} = \{(a, b] \in \mathbb{R} ; a < b\}$. Show that $\mathcal{B}$ is indeed a basis for some topology $\mathcal{O}$ on $\mathbb{R}$. Show that $\mathcal{O}$ is separable, but it does not have a countable basis. Show that $\mathcal{O}$ is not coming from a metric topology (i.e. not metrizable).

C2. Fundamental system of neighborhoods.

Recall what is a neighborhood of a point x or a subset A in some topological space X . A fundamental system of neighborhoods of x (or of a subset A) is a collection $\mathcal{P}$ of neighborhood of x (or of the subset A) such that every neighborhood of x (or of A) contains some element of $\mathcal{P}$.

Show that in a metric space X , with any r_n tending to 0, $B(x, r_n)$ is a fundamental system of neighborhoods of x . But if A is a subset and $V_\epsilon(A) = \{x \in X ; d(x, A) < \epsilon\}$,

$V_\epsilon(A)$ is not necessarily a fundamental system of neighborhoods of A . For example, consider $A = \{xy = 1, x > 0\} \subset \mathbb{R}^2$ and $U = \{x, y > 0\} \subset \mathbb{R}^2$.

C3. Hausdorff condition.

Recall what is a Hausdorff space, or T_2 space. Show that a metric space is always Hausdorff.

Show that every Zariski open set of $\mathbb{A}^n(\mathbb{R})$ (or $\mathbb{R}^n$) is open dense for the usual topology of $\mathbb{R}^n$. Show that the Zariski topology of $\mathbb{A}^n(\mathbb{R})$ is not Hausdorff for any $n > 0$.

The first Colle problem of this week has an example of a non-metrizable Hausdorff space.

C4. Normal space.

A topological space X is called normal if for any closed set F contained in an open set V , there is some open set U such that $F \subset U \subset \overline{U} \subset V$. Show that this is equivalent to saying that X is Hausdorff, and that any closed sets F, F' of X has disjoint neighborhoods. Show that a metric space is normal (you can first show that the function $f : x \rightarrow d(x, F)$ is continuous if F is closed).

Continuing in this direction one can learn about the Urysohn extension theorems.

C5. Product topology.

(...)

C6. Mixed barbecue.

Consider the set $[0, 1] \subset \mathbb{R}$ and two other elements $\{0_-, 0_+\}$ not in $[0, 1]$. Let $X = \{0_-, 0_+\} \cup (0, 1]$, and consider $\mathcal{B}(x) = \{(x - \frac{1}{n}, x + \frac{1}{n}) \cap (0, 1] ; n \in \mathbb{Z}_{>0}\}$ for $x \in (0, 1]$ and $\mathcal{B}(0_\pm) = \{\{0_\pm\} \cup (0, \frac{1}{n}) ; n \in \mathbb{Z}_{>0}\}$. Show that $\mathcal{B} = \cup_{x \in X} \mathcal{B}(x)$ is a basis for some topology on X . Show that this topology is not Hausdorff. Show that 0_+ does not have a fundamental system of closed neighborhoods.

C7. Universal property.

Let X be a topological space and Y a subset of X with the natural inclusion map $\iota : Y \rightarrow X$. Show that the subspace topology on Y is such that for any topological space Z and $f : Z \rightarrow Y$, the map f is continuous if and only if $\iota \circ f : Z \rightarrow X$ is continuous.

3

Compactness

3.1 Warm-ups

Q1. Show that $\{0, 1\}^{[0,1]}$ with the usual product topology is not sequentially compact.

Q2. We say that a family of sets has the finite intersection property (FIP) if every finite subfamily has non-empty intersection. Show that the compactness of a space X is equivalent to the following: every family $\mathcal{F}$ of closed subsets in X with the finite intersection property has $\bigcap_{F \in \mathcal{F}} F \neq \emptyset$.

Q3. Suppose that A, B are disjoint closed sets in a metric space (X, d) and that A is compact. Show that $d(A, B) > 0$.

Q4. Suppose (X, d) is a metric space with no isolated points. Prove that X is compact if and only if $d(A, B) > 0$ for any pair of closed disjoint subsets A, B of X .

Q5. Consider $\mathbb{C}$ and a point at infinity $\{\infty\}$. We denote this space by $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ and define a topology on it by declaring that, for $U \subset \widehat{\mathbb{C}}$:

- if $\infty \notin U$, then U is open if and only if it is open in the usual topology of $\mathbb{C}$;
- if $\infty \in U$, then U is open if and only if the complement of $U \cap \mathbb{C}$ in $\mathbb{C}$ is compact in $\mathbb{C}$.

Show that the space $\widehat{\mathbb{C}}$ with the above-defined topology is compact.

This is called the Alexandrov one-point compactification procedure, and $\widehat{\mathbb{C}}$ is called the Riemann sphere (or the extended complex plane). It is advisable to search for a short picture or video before solving this exercise.

3.2 Main exercise 1: the Hausdorff distance

Let (X, d) be a metric space. We want to define a distance between any two (in fact, compact and we will see why below) subsets A, B of X .

Q1. Show that the naive definition $d(A, B) = \inf_{a \in A, b \in B} d(a, b)$ is not a distance in general.

We now introduce the ϵ -neighborhood of a set $A \subset X$ defined as $U_\epsilon(A) = \{x \in X ; d(x, A) < \epsilon\}$. Recall that the distance between a point x and a set A is $d(x, A) = \inf_{a \in A} d(x, a)$.

Q2. Show that for $r_1, r_2 > 0$ and any $A \subset X$, we have $U_{r_1}(U_{r_2}(A)) \subset U_{r_1+r_2}(A)$, but the converse is not true in general.

We now define the Hausdorff distance between subsets $A, B \subset X$ as $d(A, B) = \inf\{r > 0 ; A \subset U_r(B), B \subset U_r(A)\}$.

Q3. Show that the Hausdorff distance is a pseudo-distance, i.e. it satisfies all the conditions for a distance except that $d(A, B) = 0$ does not necessarily imply that $A = B$.

Q4. In fact, show that for all $A \subset X$, $d(A, \overline{A}) = 0$. If A, B are closed, then $d(A, B) = 0$ implies $A = B$.

Q5. Show that if we restrict to the collection $\mathcal{K}(X)$ of compact subsets $A \subset X$, then the Hausdorff distance is an honest distance on $\mathcal{K}(X)$.

3.3 Main exercise 2: the de Bruijn-Erdős theorem

An unoriented graph G is a pair (V, E) with a set of vertices V and a set of edges $E \subset V \times V$. We allow multi-sets, i.e. we allow several edges between two vertices $x, y \in V$.

A graph is finite if its set of edges E is finite.

A subgraph $\tilde{G} = (\tilde{V}, \tilde{E}) \subset G$ is such that $\tilde{V} = V$ and $\tilde{E} \subset E$. Here we impose that the set of vertices are the same and we only “delete” edges.

A coloring of G by k colors is a map $f : V \rightarrow \{1, \dots, k\}$ such that for any edge $(x, y) \in E$, $f(x) \neq f(y)$, i.e. any edge in G has extremities of different colors.

Q1. Endow $\{1, \dots, k\}$ with the discrete topology. Show that $\{1, \dots, k\}$ is compact in this topology.

Q2. Use Tychonoff's theorem to show that the set of maps from $V \rightarrow \{1, \dots, k\}$ is compact. We denote this set by X .

For any subset $F \subset E$, we define $X_F = \{f : V \rightarrow \{1, \dots, k\} ; f \text{ is a coloring of } (V, F)\}$.

Q3. If $F = \{(x, y)\}$ has only one edge, show that X_F is closed. You can write this as a finite union of closed sets.

Q4. Show that for any $F \subset E$, show that X_F is closed. You can write this as an intersection of the closed sets defined in the previous question.

We now suppose that we have a graph G such that any finite subgraph of G is k -colorable, i.e. $X_F \neq \emptyset$ if F is finite.

Consider thus the following collection of non-empty closed sets $\mathcal{F} = \{X_F ; F \subset E \text{ is finite}\}$.

Q5. Show that any finite intersection of elements in $\mathcal{F}$ is non-empty.

Q6. Use the compactness of X to show that the intersection of all elements in $\mathcal{F}$ is non-empty.

Q7. Prove the theorem of de Bruijn-Erdős: if any finite subgraph of a graph G is k -colorable, then G is colorable. This works even in the case where G has uncountably many vertices.

3.4 A little more: metric space exercises

Q1. Let X be a compact metric space and $\{u_n\}_{n \geq 0}$ a sequence of X with $d(u_n, u_{n+1}) \rightarrow 0$. Show that the set of accumulation points of $\{u_n\}$ is compact, then show that it is connected.

Q2. Suppose X, Y are metric spaces and $f : X \rightarrow Y$. Show that f is continuous if and only if the restriction of f to any compact set $K \subset X$ is continuous.

Q3. Let (X, d) be a metric space and $f : X \rightarrow X$ be an isometry, i.e. $d(x, y) = d(f(x), f(y))$ for all $x, y \in X$. Show that if X is compact, then f is bijective and is a homeomorphism. You can choose $a \notin f(X)$ and look at $d(f^n(a), f^m(a))$.

3.5 Colles

C1. Simple connectedness.

Consider the plane $\mathbb{R}^2$. A plane domain is an open, path connected subset of $\mathbb{R}^2$. A plane domain is called simply connected if its complement is connected. If the complement of a plane domain consists of n connected components, then G is called n -tuply connected.

Show that a component F of the complement of a plane domain G is closed. Show that the complement of F is a simply connected domain. Show that the boundary of a n -tuply connected domain consists of n connected components. (For the last question, you need the following lemma: [...])

Show that a n -tuply connected domain can only be continuously mapped to another n -tuply connected domain. Is the converse true (that there must exist a homeomorphism between two n -tuply connected domains)? Also Google Ahlfors' five islands theorem.

C2. Review of a proof.

Recall the definition of path-connectedness. Recall the definition of the deleted comb space, and recall the proof of its non path-connectedness. How many path-connected components there are in the deleted comb space?

What if we now allow a path to be continuously parametrized by a function $f : [0, \infty) \rightarrow X$? Does the same proof still hold? What about the other proof proposed in class with compactness?

C3. Local properties.

Let X be a Hausdorff space. Show that X is locally compact, i.e. every point of X has a compact basis of neighborhood, if and only if every point of X has a compact neighborhood.

C4. ExamA, exo1

C5. ExamB, exo2

C6. Dimension and compactness.

Show that the unit ball of any normed $\mathbb{R}$ -vector space is compact if and only if the dimension of this vector space is finite.

C7. Dilatation of a compact.

Let X be a compact subspace of some normed vector space (think of $\mathbb{R}^n$). Suppose that $f : X \rightarrow X$ is a dilatation, i.e. $\forall x, y \in X, |f(y) - f(x)| \geq |y - x|$.

Show that f is an isometry (hint: show that for all $a \in X$, the sequence $f^n(a)$ comes back infinitely often near a). Show that $f : X \rightarrow X$ is a bijection.

Show that a surjective 1-Lipschitz function $g : X \rightarrow X$ is an isometry.

C8. Fixed point theorem in a compact space.

Let K be a compact subspace of some normed vector space (think of $\mathbb{R}^n$). Suppose that $f : K \rightarrow K$ is strictly contracting, i.e. $\forall x, y \in K, x \neq y, |f(y) - f(x)| < |y - x|$.

Show that f has a unique fixed point in K . You can consider the function $g : x \mapsto |f(x) - x|$. Show that for all $a \in K$, the sequence $f^n(a)$ converges to this unique fixed point.

C9. Two subsequential limits.

Let K be a compact subspace of some normed vector space (think of $\mathbb{R}^n$). Show that a sequence of K converges in K if and only if this sequence has one unique subsequential limit.

Let $f : K \rightarrow K$ be a continuous map and consider the sequence $f^n(x)$ for some $x \in K$. Show that if this sequence has exactly two subsequential limits z_0, z_1 , then for any neighborhoods V_0, V_1 of respectively z_0, z_1 , the sequence is eventually in $V_0 \cup V_1$.

Show that $f(z_0) = z_1, f(z_1) = z_0$ and that the subsequences $f^{2k}(x)$ and $f^{2k+1}(x)$ converge, one to z_0 and the other to z_1 .

4

Completeness and Baire Category

4.1 Warm-ups

Q1. Show that in the product topology on $\prod_{i \in I} X_i$, the projection maps $p_k : \prod_{i \in I} X_i \rightarrow X_k$ are all open (i.e. they map open sets to open sets). Show that they are not closed in general.

Show that the converse to the statement on openness is not true: if $W \subset \prod_{i \in I} X_i$ such that all $p_i(W)$ are open, it does not imply that W is open.

To find counter-examples, you can consider $\mathbb{R}^2$ with some points removed.

Q2. Suppose $E = \mathbb{R}^n$ with the usual topology. If K is a compact of E and F is a closed set of E , show that there exists $x \in K$ and $y \in F$ such that $d(K, F) = d(x, y)$.

Show that this is not true if we replace E by l^∞ . You can consider the set $F = \{(1 + 2^{-n})e_n\}_{n \geq 0}$, where e_n is the sequence with 1 at position n and 0 elsewhere.

Q3. We use the Baire category theorem to construct a dense family of functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which are continuous but nowhere derivable. You are allowed to find hints on the internet for this exercise.

Q4. Take two metric spaces E, F and $f : E \rightarrow F$ an injective function. Show that f is continuous if and only if for any compact $K \subset E$, $f(K)$ is compact in F . Show that this is not necessarily true if f is not injective.

4.2 Main exercise 1: two convergence arguments

The first is the “diagonal sequence trick”. Take $f_n(m)$ a sequence of functions on the positive integers that is uniformly bounded, i.e. $\exists C, \forall n, m, |f_n(m)| \leq C$.

Q1. Show that the sequence $\{f_n(1)\}_{n \geq 0}$ has a convergent subsequence $\{f_{n_1(i)}(1)\}_{i \geq 0}$.

Q2. Show that you can find successively subsequences $\{f_{n_k(i)}(m)\}_{i \geq 0}$ such that $f_{n_{k+1}(i)}$ is a subsequence of $f_{n_k(i)}$ and that $f_{n_k(i)}(k)$ converges as $i \rightarrow \infty$.

Q3. Show that there is a subsequence $f_{n_i}(m)$ so that for each m , $f_{n_i}(m)$ converges as $i \rightarrow \infty$. You can consider the sequence $n_k(k)$.

The second is the “ $\epsilon/3$ trick”. We review the proof that the uniform limit of a sequence of continuous functions is continuous. Let $C[a, b]$ be the continuous real functions on $[a, b]$ with the metric $d(f, g) = \sup_{x \in [a, b]} |f(x) - g(x)|$, induced by the norm $\|f\|_\infty = d(f, 0)$.

Q4. Let f_n be a Cauchy sequence for the $\|\cdot\|_\infty$ norm. Show that for each $x \in [a, b]$, $f_n(x)$ converges to some $f(x) \in \mathbb{R}$.

Suppose that f is the limit of a sequence of continuous functions f_n . To show continuity of f , we fix $x \in [a, b]$ and $\epsilon > 0$ and in the following exercise, we find $\delta > 0$ such that $|x - y| < \delta$ implies $|f(x) - f(y)| < \epsilon$.

Q5. Show that you can pick n so that $\|f_n - f\|_\infty < \epsilon/3$. Show that you can pick $\delta > 0$ so that $|x - y| < \delta$ implies $|f_n(x) - f_n(y)| < \epsilon/3$. Show that f is continuous.

Q6. Show that $f_n \rightarrow f$ in $C[a, b]$, so that $C[a, b]$ is complete.

4.3 Main exercise 2: Banach’s fixed point theorem

Let (X, d) be a non-empty complete metric space and suppose that a map $f : X \rightarrow X$ is strictly contracting, i.e.

$$\exists 0 < k < 1, \forall x, y \in X, d(f(x), f(y)) \leq k d(x, y).$$

Q1. Show that f has a unique fixed point (i.e. $x \in X$ such that $f(x) = x$) in X .

The above exercise is Banach’s fixed point theorem. Below is a classical application of this theorem.

Let $(E, |\cdot|)$ be a complete normed vector space over a field $\mathbb{K}$ and $f : E \rightarrow E$ is a continuous linear map.

Q2. Show that there exists $K > 0$ such that $\forall x, y \in E, |f(x) - f(y)| \leq K|x - y|$.

Q3. Show that for any given $\lambda \in E, |\lambda| > K$, the map with fixed $a \in E$ defined by $g_a(x) = \frac{a - f(x)}{\lambda}$ is contracting. Use Banach's fixed point theorem to deduce that g_a has a unique fixed point $J_a \in E$.

Q4. Show that $|J(a) - J(b)| \leq \frac{1}{|\lambda| - K}|a - b|$.

Q5. Conclude the following. For any continuous linear map $f : E \rightarrow E$, there exists $K > 0$ such that for any $\lambda \in E, |\lambda| > K$, for any $a \in E$, the equation $f(x) - \lambda x = a$ on the variable $x \in E$ has a unique solution $J(a)$, and the map $a \rightarrow J(a)$ is continuous.

4.4 A little more: metric space exercises

Q1. The following exercise is a baby version of the Arzelà-Ascoli theorem.

Let $M \geq 0$ and $\{f_n\}_{n \geq 0}$ a sequence of functions on $[0, 1]$, of class C^1 , such that $|f(x)| + |f'(x)| \leq M$ for all $x \in [0, 1]$. We want to show that there is a subsequence of $\{f_n\}$ that converges uniformly on $[0, 1]$.

1. Use the diagonal argument to show that there is a subsequence of $\{f_n\}$ that converges pointwise on rational numbers of $[0, 1]$.
2. Denote the previous subsequence as $\{g_n\}$. Use the $\epsilon/3$ argument to show that $\{g_n\}$ converges pointwise on all points of $[0, 1]$.
3. Denote the (pointwise) limit in the previous question as g . Use again the $\epsilon/3$ argument to show that g is actually the uniform limit of $\{g_n\}$ on $[0, 1]$.

Q2. Let E be a compact metric space and $f : E \rightarrow E$ be a continuous map with $d(f(x), f(y)) \geq d(x, y)$ for all $x, y \in E$.

1. Show that f is actually an isometry, i.e. $d(f(x), f(y)) = d(x, y)$ for all $x, y \in E$.
2. Show that the f above is actually bijective. You can consider subsequences of $(x_n, y_n) = (f^n(x), f^n(y))$.
3. Finally, show that if $g : E \rightarrow E$ is continuous bijective and $d(g(x), g(y)) \leq d(x, y)$ for all $x, y \in E$, then g is again an isometry.

Q3. Use the Baire category theorem to show that if a sequence of continuous functions $f_n : [0, 1] \rightarrow \mathbb{R}$ converges pointwise to $f : [0, 1] \rightarrow \mathbb{R}$, then f cannot be everywhere discontinuous.

Q4. Show that a normed vector space is complete if and only if all absolutely convergent series are convergent.

4.5 Colles

C1. Sequence and metric space.

Show that the limit of a sequence in a Hausdorff space is unique. Show that in a metric space, the limit points of a sequence are the subsequential limits. What happens if the space is not a metric space?

C2. Sequence, metric space and continuity.

Show that for $f : X \rightarrow Y$ between metric spaces, the function is continuous if and only if for any convergent sequence in X , its image by f is a convergent sequence in Y . Which parts of the proof use the metric space structure? Can you relax some assumptions?

C3. Connectedness and continuous function.

Show that a space X is connected if and only if all continuous functions $f : X \rightarrow \{0, 1\}$ with the discrete topology at the target space are constant.

Consider two connected metric spaces A, B , and let X, Y be proper subsets of respectively A, B . Show that $C = (A \times B) \setminus (X \times Y)$ is a connected subset of $A \times B$.

C4. Connectedness and frontier.

Let E be a connected metric space and F a closed set of E . Show that if the frontier of F is connected, then F is connected. You can first show that the restriction of a continuous function is continuous. Give a counterexample in the case where F is not supposed closed.

C5. Chained space.

We say that a metric space X is ϵ -chained if for all $a, b \in X$, we can find a finite sequence of points $x_i \in X$ with $x_0 = a, x_n = b$ and $|x_i - x_{i+1}| < \epsilon$ for all $0 \leq i \leq n-1$. We say that X is well-chained if X is ϵ -chained for all $\epsilon > 0$.

Show that if X is connected, then X is well-chained. Show that if X is compact and well-chained, then X is connected. Find an example of a well-chained space which is not connected.

C6. Fixed point and connectedness.

Let X be a metric space. Suppose that all continuous functions $f : X \rightarrow X$ have a fixed point. Show that X is connected.

Now suppose that X is compact and connected. Is it true that all functions $f : X \rightarrow X$ has a fixed point?

C7. Exactly one accumulation point.

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuous, and suppose that some sequence $f^n(x_0)$ has exactly one accumulation point. Show that this sequence converges.

C8. Isometry and compactity (and Lebesgue number).

Consider a non-empty compact space $K \subset \mathbb{R}^n$ and $\epsilon > 0$. We say that $A \subset K$ is ϵ -separated if for all $x, y \in A$, $x \neq y$ implies $|x - y| > \epsilon$. Show that there is some finite integer $M(\epsilon)$ such that all subsets $A \subset K$ that is ϵ -separated has at most cardinal $M(\epsilon)$.

Suppose that $f : K \rightarrow K$ is an isometry. Show that f is surjective.

C9. Neighborhood of a dense set.

Is it true that any neighborhood of a dense set in X covers X ? You can consider the example of $\cup B(\frac{p}{q}, \frac{1}{q^k})$ for $\frac{p}{q} \in \mathbb{Q}$ and k large enough.

C10. Sum of closed sets.

Let $E = \mathbb{R}^n$ and $A, B \subset E$. If A is compact and B closed, show that $A + B$ is closed. If A, B are closed, show that $A + B$ is not necessarily closed.

5

Constructions of Topology

Nearly all the exercises are taken from the lecture notes of Paulin.

Q1. Let X be a topological space. Show that the following are equivalent:

1. X has the discrete topology;
2. X has the weak (final) topology induced by the family of $\{x\}_{x \in X}$;
3. The canonical map from the disjoint sum $\coprod_{x \in X} \{x\}$ to X is a homeomorphism.

Q2. Suppose that if X_i are topological spaces with $X_i \subset X_{i+1}$ with $X = \cup_{i=1}^{\infty} X_i$. Suppose that for each i , X_i is closed in X_{i+1} and that the topology of X_i is the subspace topology induced by the topology on X_{i+1} . Then for each i , the topology on X_i coincides with the subspace topology induced by the weak (final) topology on X , where the weak topology on X is induced by the family X_i . Furthermore, show that each X_i is closed in X .

Q3. Let X be a topological space and $\pi : X \rightarrow Y$ a quotient map. Let Y be equipped with the quotient topology induced by π .

1. Show that if X is connected, then Y is connected.
2. Show that if X is path connected, then Y is path connected.
3. Show that if π is an open map and X is locally connected (resp. path connected), then Y is connected (resp. path connected).

Q4. (Equivalence relation generated by a binary relation) Let X be a set and $\mathcal{R}$ a relation on $X \times X$, not necessarily an equivalence relation. In general, it is just a

subset of $\mathcal{R} \subset X \times X$ where $x\mathcal{R}y$ if and only if $(x, y) \in \mathcal{R}$. Notice that $x\mathcal{R}y$ does not imply $y\mathcal{R}x$ in general.

Show that the following definition is well-posed: there exists a unique, smallest equivalence relation $\sim$ containing $\mathcal{R}$ (in the sense of subsets of $X \times X$), called the equivalence relation generated by the relation $\mathcal{R}$. You can first show that the intersection of a family of equivalence relations is still an equivalence relation.

Q5. (Quotient distance of a pseudo-distance) Recall that a pseudo-distance d is almost a distance without the requirement that $d(x, y) = 0$ implies $x = y$. A pseudo-distance space (X, d) also has a topology with the basis of open balls $B(x, \epsilon)$ in the pseudo-distance d .

We can define an equivalence relation $\sim$ by that $x \sim y$ if and only if $d(x, y) = 0$. Show that this is indeed an equivalence relation, and that the quotient space $X/\sim$ can be equipped with a distance d' such that $d'([x], [y]) = d(x, y)$ for any x, y . Check that the above definition of d' is well-posed.

Show that the topological spaces $(X/\sim, d')$ (as a metric space) and $X/\sim$ as a quotient space are the same. Show that if F is any Hausdorff space and $f : X \rightarrow F$ is continuous, then f factorizes through $X/\sim$, i.e. there exists $f' : X/\sim \rightarrow F$ such that $f = f' \circ \pi$. We say that $X/\sim$ is the largest Hausdorff quotient of (X, d) .

Q6. Show that a real or complex topological vector space is locally convex if and only if its topology is defined by a family of semi-norms. Show that in a real or complex topological vector space, the origin 0 has a countable basis of convex neighborhood if and only if the topology is defined by a countable family of semi-norms.

Q7. Let $\pi : \mathbb{R}^2 \rightarrow \mathbb{T}^2$ be the canonical projection and $\alpha \in \mathbb{R} \cup \{\infty\}$. Let D be a straight line of slope α in $\mathbb{R}^2$. Show that if $\alpha \in \mathbb{Q} \cup \{\infty\}$, then $\pi(D)$ is homeomorphic to a circle. Show that if $\alpha \notin \mathbb{Q} \cup \{\infty\}$, then $\pi(D)$ is dense in $\mathbb{T}^2$ and π restricted to D is a continuous bijection between D and $\pi(D)$ (but is it a homeomorphism?).

Q8. Let $\pi : \mathbb{R}^2 \rightarrow \mathbb{T}^2$ be the canonical projection and $\alpha \in \mathbb{R} \cup \{\infty\}$. Let $\sim$ be an equivalence relation on $\mathbb{T}^2$ such that $x' \sim y'$ in $\mathbb{T}^2$ if and only if there are points $x, y \in \mathbb{R}^2$ and a straight line $\{x, y\} \subset D \subset \mathbb{R}^2$ of slope α such that $\pi(x) = x'$ and $\pi(y) = y'$. Show that $\sim$ is indeed an equivalence relation. Show that the quotient space $\mathbb{T}^2/\sim$ is Hausdorff if and only if $\alpha \in \mathbb{Q}$, in which case show that the quotient space is homeomorphic to a circle. If $\alpha \notin \mathbb{Q}$, show that the quotient space $\mathbb{T}^2/\sim$ comes with a trivial topology.

Q9. (The cone) Consider the cone space CX over a topological space X defined in the course. Show that the map from X to CX defined as $x \mapsto [(x, 0)]$ is a homeomorphism onto its image, so that X can be identified as a subset of CX . Verify that if $f : X \rightarrow Y$ is continuous, then $Cf : CX \rightarrow CY$ defined by $[(x, t)] \mapsto [(f(x), t)]$ is also continuous.

Q10. (The suspension) Consider the sphere $\mathbb{S}^n = \{x_1^2 + x_2^2 + \cdots + x_{n+1}^2 = 1\} \subset \mathbb{R}^{n+1}$. Show that the suspension $S(\mathbb{S}^n)$ is homeomorphic to $\mathbb{S}^{n+1}$.

Q11. (The crushing) Show that if $A \subset X$ is open (or closed), then the projection $X \rightarrow X/\langle A \rangle$ restricted to $X - A$ is a homeomorphism. Show that $CX/\langle X \rangle$ and SX are homeomorphic (here we identify X with $\pi(X \times \{0\}) \subset CX$).

Q12. Let X be a compact Hausdorff space. Show that the cone space CX is homeomorphic to the one-point compactification of $X \times [0, 1)$. If $A \subset X$ is closed, show that $X/\langle A \rangle$ is homeomorphic to the one-point compactification of $X \setminus A$.

Q13. (The n -dimensional torus) Recall the n -dimensional torus defined in the course and the map

$$\bar{\phi} : [(t_1, \dots, t_n)] \mapsto (e^{i2\pi t_1}, \dots, e^{i2\pi t_n}).$$

1. Show that $\bar{\phi}$ is continuous.
2. Show that $\mathbb{T}^n$ is Hausdorff.
3. Show that $\bar{\phi}$ is a homeomorphism.

Q14. Let E be a topological vector space over $\mathbb{K}$. Let C be a convex neighborhood of 0 and define the gauge function $\|\cdot\|_C : E \rightarrow [0, \infty)$ by the formula $\|x\|_C = \inf_{t>0} \{ \frac{1}{t}x \in C \}$. If $\{C_\alpha\}_{\alpha \in \mathcal{A}}$ is a family of convex subsets in E such that their intersection is a convex neighborhood of 0, then show that

$$\|\cdot\|_{\cap_{\alpha \in \mathcal{A}} C_\alpha} = \sup_{\alpha \in \mathcal{A}} \|\cdot\|_{C_\alpha}.$$

Q15. Let E, F, G be topological vector spaces over a normed field $\mathbb{K}$. For all $f \in \mathcal{L}(E, F)$, we can define the right composition by f from $\mathcal{L}(F, G)$ to $\mathcal{L}(E, G)$ by $g \mapsto g \circ f$. Show that the operator norm of the right composition by f is less or equal to the operator norm of f . Show that the same conclusion holds for the left composition.

Show that the map from E to $\mathcal{L}(\mathbb{K}, E)$ defined by $x \mapsto \{\varphi_x : t \mapsto tx\}$ is an isometric linear isomorphism.

Q16. Show that if E is an infinite dimensional topological vector space, then the strong topology and the weak topology on E are not the same. Calculate the interior of the open unit ball of E in the weak topology. Calculate the closure of the unit sphere of E in the weak topology.

Q17. Show that the weak-star topology is always Hausdorff.

Q18. Let E be a normed topological vector space over $\mathbb{R}$ and $D \subset E$. Let E^* be the topological dual of E , with the initial topology induced by the family of linear forms $\{\varphi_x : E^* \rightarrow \mathbb{R}\}_{x \in D}$. Show that this topology is coarser than the weak-star topology on E^* . Show that if D is dense in E , then on all bounded sets in E^* (for the dual norm), the restriction of the above-defined topology and the weak-star topology are the same.

5.1 Colles

C1. Semi-norm topology.

C2. Gauge function of a convex set.

C3. Zariski-compact does not imply Zariski-closed.

C4. One-point compactification.

See Exercise 3.1.Q5 of Week 3.

C5. Completion.

Suppose that (M, d) is an incomplete metric space. Consider all Cauchy sequences $\{x_n\}$ with the equivalence relation that $\{x_n\} \sim \{y_n\}$ if $\lim_{n \rightarrow \infty} d(x_n, y_n) = 0$. Let $\tilde{M}$ be the family of equivalence classes of Cauchy sequences, with the distance induced by $\lim_{n \rightarrow \infty} d(x_n, y_n)$. Show that $\tilde{M}$ is complete, and that M is dense in $\tilde{M}$ under the inclusion map $x \in M \mapsto (x, x, \dots) \in \tilde{M}$.

C6. Baire category theorem.

Recall the statement of the Baire category theorem in a complete metric space. Show that the theorem is also true in a locally compact Hausdorff space.

C7. Dense convex set.

Show that a dense convex set in $\mathbb{R}^n$ is $\mathbb{R}^n$. Use the Weierstrass theorem on the density of polynomial functions to show that this is wrong in infinite dimension.

C8. Open balls in a Banach space.

Suppose that $X = \mathbb{R}^n$. Suppose that Ω is open bounded and non-empty in X , and that for all $x, y \in \Omega$, there exists some open ball contained in Ω containing x, y . Show that Ω is itself an open ball of $\mathbb{R}^n$. Find a counter-example when Ω is not supposed bounded. Same exercise when X is any Banach space.

C9. Example of Banach space I.

C10. Example of Banach space II.

C11. Example of a non-Banach space.

Consider the space $L^p([0, 1])$ for $0 < p < 1$, i.e.

C12. Equivalent of continuity and boundedness for linear transformations.

Let T be a linear transformation between two normed linear spaces. The following are equivalent:

1. T is continuous at one point;
2. T is continuous at all points;
3. T is bounded.

C13. Choice of a norm.

Let Q be a polynomial of $\mathbb{R}[X]$. Construct a norm over $\mathbb{R}[X]$ such that the sequence $\{X^n\}_{n \geq 0}$ converges to Q in this norm.

C14. Continuity of the composition.

Consider the space $E = C^0([a, b], \mathbb{R})$ with the $\|\cdot\|_\infty$ norm. Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Show that the map $f \in E \mapsto \varphi \circ f \in E$ is continuous.

C15. Not Baire.

Let E be a normed vector space that is not Baire. Show that E is a countable union of nowhere dense closed sets.

C16. Normed vector space proposition.

C17. Quotient norm.

C18. Connected frontier.

Suppose that (E, d) is a connected metric space. Let F be closed in E such that ∂F is connected. Show that F is connected.

6

More Colles exercises

C1. Complement of a compact set.

Show that the complement of a compact set in an infinite dimensional vector space is path-connected. Is this true for a finite dimensional vector space?

C2. Global extremum. (HSH)

Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous with $n \geq 2$. We suppose that $f^{-1}(a)$ is compact for all $a \in \mathbb{R}$. Show that f has a global extremum (either supremum or infimum). What happens when $n = 1$?

C3. Hyperplan and closedness. (LZY)

Let E be a normed vector space and H a hyperplane, that is H is of codimension 1 in E . Show that H is the kernel of some linear form $\varphi : E \rightarrow \mathbb{R}$. Show that H is closed if and only if φ is continuous.

Proof. The first question is a classic in linear algebra.

One direction for the second question is clear using pullback of continuous map. For the other direction, suppose that H is closed and φ is not continuous: φ is thus not bounded, and we can find a sequence of $|x_n| = 1$ such that $|\varphi(x_n)| \rightarrow \infty$. Let $u \in E$ and define $u_n \in H$ by projecting u onto H in the direction x_n : concretely, define $u_n = u - \frac{\varphi(u)}{\varphi(x_n)}x_n$ (check that $\varphi(u_n) = 0$). Then $|u - u_n| = \frac{|\varphi(u)|}{|\varphi(x_n)|} \rightarrow 0$, so that u_n converges to u . Since H is supposed closed, we get that $u \in H$ so that $\varphi(u) = 0$. But then this shows that $\varphi \equiv 0$, which contradicts the assumption that φ is not continuous.

C4. Hyperplan and complement.

Let E be a normed vector space and H a hyperplane. Show that $E \setminus H$ is path-connected if and only if H is closed.

Proof. Let φ be a linear form on E with $H = \ker \varphi$. Then $H^- = \varphi^{-1}(\mathbb{R}_{<0})$, $H^+ = \varphi^{-1}(\mathbb{R}_{>0})$ are convex, so path-connected. By the exercise above, H is closed if and only if φ is continuous.

If H is closed, there is no continuous path from H^- to H^+ by the intermediate value theorem. Now suppose that H is not closed, so that φ is not continuous. In this case, $\overline{H} = E$ (since $\overline{H}$ is a linear space that is strictly larger than H when H is not closed), and H is dense in E . This is somewhat surprising since we are proving that $E \setminus H$ is path-connected. It suffices to continuously join two elements $-e \in H^-$ and $e \in H^+$, say with $\varphi(e) = 1$. But $\varphi^{-1}(1) = e + H$ is dense in E , so there exists a sequence $\{x_n\} \in e + H$ which converges to $-e$, and we can suppose that $x_1 = e$. Now we define the path $\gamma : [0, 1] \rightarrow E$ in the following way: γ connects x_{k+1} to x_k using a straight segment from time $\frac{1}{k+1}$ to $\frac{1}{k}$, and $\gamma(0) = -e$. If we can show that γ is continuous at 0, we are done since γ is continuous on $[0, 1]$ and $\gamma([0, 1]) \subset E \setminus H$ because $\varphi(\gamma(t)) = 1$ for all $t \in (0, 1]$ and $\varphi(\gamma(0)) = -1$.

We now show that γ is continuous at 0: but this is basically just the assumption that $\{x_n\}$ converges to $-e$ (and some convexity argument). This completes the proof.

C5. Easy consequence of the open mapping theorem.

C6. Easy consequence of the Banach-Steinhaus theorem.

C7. Easy consequence of the Banach-Schauder theorem.

C8. Baire category exercise I.

Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ be a continuous map. Suppose that for all $x \geq 0$, the sequence $\{f(nx)\}_{n \geq 0}$ converges to 0. Show that $\lim_{x \rightarrow \infty} f(x) = 0$.

This exercise is sometimes known as Croft's lemma.

C9. Baire category exercise II.

Let E be a $\mathbb{R}$ -vector space of dimension $1 \leq n < \infty$, and suppose that $\{F_k\}_{k \in \mathbb{N}}$ is a sequence of p -dimensional subvector spaces of E .

Show that $\cup_{k \in \mathbb{N}} F_k \neq E$. Show that there exists some subspace W of E such that for all $k \in \mathbb{N}$, $W \oplus F_k = E$.

C10. Baire category exercise III.

Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ be a continuous map. Suppose that for all $\lambda \geq 0$, $\lim_{x \rightarrow \infty} f(x + \lambda) - f(x) = 0$. Show that for all $[a, b] \subset \mathbb{R}_+$, this convergence is uniform in $\lambda \in [a, b]$.

C11. Baire category exercise IV.

Consider the space ℓ^q with $1 < q < \infty$. Then $\ell^q \neq \cup_{1 < p < q} \ell^p$.

C12. Quotient topology.

Check the equivalence of three definitions of the quotient topology.

C13. Théorème de l'arbre et de l'écorce.

Show this theorem from the course, but with the functional method.

7

Revision exercises

Here is a list of classical exercises from French oral exams. They are exclusively metric space exercises taken from the book “Les maths en tête” of Xavier Gourdon.

7.1 Compactness

In a metric space topology course one usually starts with compactness...

Let (E, d) be a metric space and $f : E \rightarrow \mathbb{R}$ a continuous map. Suppose that f restricted to some compact set $K \subset E$ is injective and $\forall x \in K, \exists \epsilon > 0$, such that f restricted to $B(x, \epsilon)$ is injective. Show that f is injective on some open set Ω containing K .

You can try to “enlarge” K a little by consider its $\frac{1}{n}$ -flattening, and use an argument by contradiction.

The following exercise is known as the Riesz theorem: the closed unit ball of an infinite dimensional normed vector space E is not compact.

Let E be a real normed vector space of infinite dimension. Then the closed unit ball of E cannot be covered by finitely many open balls of radius 1.

If it can be covered by open unit balls with center $\{x_i\}_{1 \leq i \leq n}$, consider $F = \text{span}(x_1, \dots, x_n)$ and use some element $y \in E \setminus F$ to construct some x_0 in the unit sphere exactly at distance 1 to F .

Finally, a classical theorem that we didn't have time to discuss in the main course. It is known under the name of the extension theorem, attributed to Tietze or Urysohn.

If (E, d) is a metric space and $A \subset E$ is a closed set with a function $f : A \rightarrow \mathbb{R}$ which is continuous and bounded. Then there exists some continuous function $g : E \rightarrow \mathbb{R}$, coinciding with f on A , such that $\sup_{x \in E} g(x) = \sup_{x \in A} f(x)$ and $\inf_{x \in E} g(x) = \inf_{x \in A} f(x)$.

You can always suppose that $\sup_{x \in A} f(x) = 2$ and $\inf_{x \in A} f(x) = 1$, then consider the expression $\frac{1}{d(x, A)} \inf_{y \in A} (f(y)d(x, y))$ for $x \notin A$.

7.2 Connectedness

Problems about connectedness are usually simple (because the definition is simple enough), and often go by contradiction.

Suppose that (E, d) is a metric space, and that all applications $f : E \rightarrow E$ has a fixed point. Show that E is connected.

You can argue by contradiction and construct a special function.

Remember that in a connected set, a locally constant function is globally constant. (GXR)

Let $K \subset [0, 1]^2$ be closed. Suppose that for all $x \in [0, 1]$, the set $K_x = \{y \in [0, 1] ; (x, y) \in K\}$ is a non-trivial interval. Show that K is connected. Furthermore, show that there exists $x \in [0, 1]$ such that $x \in K_x$.

You can show that all continuous maps f from K to $\{0, 1\}$ is constant. Indeed, it is already constant if we fix the parameter x , i.e. $f(x, \cdot) : K_x \rightarrow \{0, 1\}$ is constant, denoted $g(x)$. You just need to show that g is locally constant.

The intermediate value theorem is a classical exercise about connectedness.

1. Let $f : [0, 1] \rightarrow [0, 1]$ be a non-decreasing function (not necessarily continuous). Show that f has a fixed point.
2. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function (not necessarily increasing). Show that f has a fixed point.
3. Let $f, g : [0, 1] \rightarrow [0, 1]$ two continuous functions with $f \circ g = g \circ f$. Show that for some $a \in [0, 1]$ we have $f(a) = g(a)$.
4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous map and suppose that $f \circ f$ has a fixed point. Show that f has a fixed point.

There are other ways to a fixed point theorem: especially in a complete space we have Banach's fixed point theorem (see exercises in previous weeks).

7.3 Completeness

Since nobody has done this exercise, it is very crucial to know the statement.

Show that a normed vector space is complete if and only if all absolutely convergent series are convergent.

The trick is to use Cauchy's criterion to construct a sequence that decreases exponentially fast, which ensures the absolute convergence.

You can also show the following result with Baire category theorem (see below).

Show that a normed vector space E with an infinite countable basis cannot be complete.

Take a normalized basis $\{e_n\}_{n \geq 1}$ of E with $\|e_n\| = 1$. Consider $F_n = \text{span}\{e_1, \dots, e_n\}$, and define $\lambda_1 = 1/3$ and $\lambda_{n+1} = \frac{1}{3}d(\lambda_n e_n, F_{n-1})$. Show that $\sum \lambda_n e_n$ converges absolutely but cannot converge.

7.4 Baire category theorem

The following exercises are about the Baire category theorem.

Show that a normed vector space E with an infinite countable basis cannot be complete.

Take a basis $\{e_n\}_{n \geq 1}$. You can consider $F_n = \text{span}\{e_1, \dots, e_n\}$ and use the Baire category theorem.

We now show that a derivative is continuous on a dense set.

Let $(E, d), (F, \delta)$ be two metric spaces with E complete. Suppose that a sequence of continuous functions $f_n : E \rightarrow F$ converges pointwise to a function $f : E \rightarrow F$.

1. For all $\epsilon > 0$ and $n \in \mathbb{N}$, define

$$F_{n,\epsilon} = \{x \in E ; \forall p \geq n, \delta(f_p(x), f_n(x)) \leq \epsilon\}.$$

Use Baire to show that Ω_ϵ , the union of the interiors of all $F_{n,\epsilon}$ for $n \in \mathbb{N}$, is dense in E .

2. Show that for all $x_0 \in \Omega_\epsilon$, there is some neighborhood V of x_0 such that for all $x \in V$, $\delta(f(x_0), f(x)) \leq 3\epsilon$.
3. Show that the set of continuity of f is a dense.
4. Show that the set of continuity of the derivative $f' : \mathbb{R} \rightarrow \mathbb{R}$ of a derivable function f is dense in $\mathbb{R}$.

8

Solutions to revision exercises

8.1 Compactness

Let (E, d) be a metric space and $f : E \rightarrow \mathbb{R}$ a continuous map. Suppose that f restricted to some compact set $K \subset E$ is injective and $\forall x \in K, \exists \epsilon > 0$, such that f restricted to $B(x, \epsilon)$ is injective. Show that f is injective on some open set Ω containing K .

Proof. Consider the $1/n$ -flattening (or $1/n$ -neighborhood of K), defined as

$$K_n = \{x \in E ; d(x, K) < 1/n\}.$$

For each integer $n \geq 1$, K_n is an open set containing K : we show that f is injective on some K_n for large enough n .

Suppose by contradiction that f is not injective for all $n \geq 1$. That is, for all $n \in \mathbb{N}$, there exists $x_n, y_n \in K_n$ with $x_n \neq y_n$ such that $f(x_n) = f(y_n)$. Then there exists $x'_n, y'_n \in K$ such that $d(x_n, x'_n) < 1/n$ and $d(y_n, y'_n) < 1/n$. Since $K \times K$ is compact, there is a convergence subsequence of $(x_{\varphi(n)}, y_{\varphi(n)})$ towards some $(x, y) \in K^2$. Also, $x_{\varphi(n)} \rightarrow x, y_{\varphi(n)} \rightarrow y$ by the triangle inequality.

Therefore by continuity of f , we have that $f(x) = f(y)$ since $f(x_{\varphi(n)}) = f(y_{\varphi(n)})$, and we get that $x = y$ since f is injective on K . But then for this common point $x = y$, $B(x, \epsilon)$ cannot be injective for any $\epsilon > 0$ since there is always some $x_{\varphi(n)} \neq y_{\varphi(n)}$ in this ball with $f(x_{\varphi(n)}) = f(y_{\varphi(n)})$.

Let E be a real normed vector space of infinite dimension. Then the closed unit ball of E cannot be covered by finitely many open balls of radius 1.

Proof. Suppose that the closed unit ball of E can be covered by open unit balls with center $\{x_i\}_{1 \leq i \leq n}$. Consider $F = \text{span}(x_1, \dots, x_n)$, which is finite-dimensional. Therefore, there exists some $y \in E \setminus F$, and denote by $x \in F$ such that $d(y, E) = d(y, x)$. The existence of x is obtained by considering $B(y, |y|) \cap E$, which is a bounded closed set in the finite dimensional normed vector space E .

Set $x_0 = \frac{y-x}{|y-x|}$ in the unit sphere of E exactly at distance 1 to F . This is because $d(x_0, F) \leq |x_0| = 1$ and for any $z \in F$, $d(x_0, z) \geq \frac{d(x, F)}{|x-y|} \geq 1$.

But then x_0 cannot be in any $B(x_i, 1)$ since its distance to F is 1.

In particular, we proved that E is not compact.

If (E, d) is a metric space and $A \subset E$ is a closed set with a function $f : A \rightarrow \mathbb{R}$ which is continuous and bounded. Then there exists some continuous function $g : E \rightarrow \mathbb{R}$, coinciding with f on A , such that $\sup_{x \in E} g(x) = \sup_{x \in A} f(x)$ and $\inf_{x \in E} g(x) = \inf_{x \in A} f(x)$.

Proof. Suppose that $\sup_{x \in A} f(x) = 2$ and $\inf_{x \in A} f(x) = 1$, then consider the function $g(x) = \frac{1}{d(x, A)} \inf_{y \in A} (f(y)d(x, y))$ for $x \notin A$, and $g(x) = f(x)$ for $x \in A$.

The only non-trivial point to verify is that g is continuous.

- If x is in the interior of A , this is ok since $g = f$.
- If x is in the complement of A , this is ok by considering the variation of g on $B(x, d(x, A)/2)$ and use triangular inequality.
- Suppose now x is on the boundary of A and let $\epsilon > 0$. Since f is continuous at y , there exists $r > 0$ such that for all $y \in B(x, r) \cap A$, $|g(y) - g(x)| = |f(y) - f(x)| < \epsilon$. Now suppose that $z \in E \setminus A$ and $|z - x| < r/4$, and show that $|g(z) - g(x)| = |g(z) - f(x)| < \epsilon$: this will show the continuity of g at x .

First, consider $C = B(z, r) \cap A$ and show that the inf for $g(z)$ is taken inside C , i.e. $\inf_{y \in A} (f(y)d(z, y)) = \inf_{y \in C} (f(y)d(z, y))$. This is because $\inf_{y \in A} (f(y)d(z, y)) \leq f(x)d(z, x) = \frac{r}{4}f(x) \leq \frac{r}{2}$, and if $y \in A \setminus C$, $f(y)d(z, y) \geq d(z, y) \geq d(x, y) - d(z, x) = \frac{3r}{4}$.

Now on C , $(f(x) - \epsilon)d(z, A) \leq \inf_{y \in C} (f(y)d(z, y)) \leq (f(x) + \epsilon)d(z, A)$, so we get $g(x) - \epsilon \leq g(z) \leq g(x) + \epsilon$ for all $z \in E \setminus A$ with $|z - x| \leq r/4$. This proves the continuity of g at the boundary of A .

8.2 Connectedness

Suppose that (E, d) is a metric space, and that all maps $f : E \rightarrow E$ have a fixed point. Show that E is connected.

Proof. We argue by contradiction: suppose that U, V is a separation of E into non-empty disjoint open sets. Then there are $u \in U, v \in V$ and the map $f(U) = \{v\}$ and $f(V) = \{u\}$ is continuous, but has no fixed point.

Let $K \subset [0, 1]^2$ be closed. Suppose that for all $x \in [0, 1]$, the set $K_x = \{y \in [0, 1] ; (x, y) \in K\}$ is a non-trivial interval. Show that K is connected. Furthermore, show that there exists $x \in [0, 1]$ such that $x \in K_x$.

Proof. We show that all continuous maps f from K to $\{0, 1\}$ is constant. Indeed, it is already constant if we fix the parameter x , i.e. $f(x, \cdot) : K_x \rightarrow \{0, 1\}$ is constant, denoted $g(x)$. We now show that g is locally constant.

Suppose otherwise, so that there exists $x_n \in [0, 1]$ converging to $x \in [0, 1]$ and $g(x_n) \neq g(x)$ for all x_n . For all x_n , take $y_n \in K_{x_n}$. The set K is compact, so that there exists a subsequence $(x_{\varphi(n)}, y_{\varphi(n)})$ converging to $(x, y) \in K$. By continuity of f on K , $g(x) = f(x, y) = \lim_{n \rightarrow \infty} f(x_{\varphi(n)}, y_{\varphi(n)}) = \lim_{n \rightarrow \infty} g(x_{\varphi(n)})$, but this contradicts with $g(x) \neq g(x_n)$ for all x_n .

The second part is a classical connectedness result.

1. Let $f : [0, 1] \rightarrow [0, 1]$ be a non-decreasing function (not necessarily continuous). Show that f has a fixed point.
2. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function (not necessarily increasing). Show that f has a fixed point.
3. Let $f, g : [0, 1] \rightarrow [0, 1]$ two continuous functions with $f \circ g = g \circ f$. Show that for some $a \in [0, 1]$ we have $f(a) = g(a)$.
4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous map and suppose that $f \circ f$ has a fixed point. Show that f has a fixed point.

Proof. • Let $A = \{x \in [0, 1] ; x \leq f(x)\}$. The set A contains 0, and let a be its supremum. Then for all $x \in A$, $x \leq f(x) \leq f(a)$ since f is non-decreasing, so $a \leq f(a)$. Therefore also $f(a) \leq f(f(a))$, so $f(a) \in A$, and $f(a) \leq a$. This shows that $f(a) = a$.

- This is classical by considering the function $g(x) = f(x) - x$.
- By the previous question, f has a non-trivial set of fixed points E . Then $g(f(x)) = g(x) = f(g(x))$ for any $x \in E$, so $g(E) \subset E$. Let $m = \inf E$ and $M = \sup E$, so the continuous function $f - g$ changes sign on $[m, M]$ and we use the previous question to conclude.
- This is quite easy: if $f(x) > x$ for all $x \in \mathbb{R}$, $f(f(x)) > f(x) > x$ so $f \circ f$ cannot have any fixed point.

8.3 Completeness

Show that a normed vector space is complete if and only if all absolutely convergent series are convergent.

Proof. First, suppose that in a normed vector space E , all absolutely convergent series are convergent. Let $\{x_n\}_{n \geq 1}$ be a Cauchy sequence and show that it converges. In fact, it is enough to show that it has a convergent subsequence (and by Cauchy-ness the original sequence converges to the same limit). Consider thus the series defined by $a_1 = x_1, a_2 = x_{\varphi(2)} - x_1, a_3 = x_{\varphi(3)} - x_{\varphi(2)}$ etc. for a subsequence $x_{\varphi(n)}$, so that the partial sum of this series is exactly $x_{\varphi(n)}$. To show that the series is convergent, it is enough to choose the subsequence, i.e. φ in such a way that the series is absolutely convergent. By Cauchy-ness, we can always choose $\varphi(n)$ so that for all $p \geq \varphi(n), |x_p - x_{\varphi(n)}| \leq 2^{-n}$. This implies absolute convergence so we are done.

Conversely, suppose that the normed vector space E is complete and let $S_n = a_1 + \dots + a_n$ be an absolutely convergent series. We should show that S_n converges. But S_n is a Cauchy sequence by absolute convergence, so it converges.

Show that a normed vector space E with an infinite countable basis cannot be complete.

Proof. Take a normalized basis $\{e_n\}_{n \geq 1}$ of E with $\|e_n\| = 1$. Consider $F_n = \text{span}\{e_1, \dots, e_n\}$, and define $\lambda_1 = 1/3$ and $\lambda_{n+1} = \frac{1}{3}d(\lambda_n e_n, F_{n-1})$. We now show that $\sum \lambda_n e_n$ converges absolutely but does not converge.

Indeed, it is quite easy to see that $|\lambda_n| \leq 3^{-n}$ by recurrence, so that the series is absolutely convergent. To see that it does not converge, suppose that it converges to x , and x must be in a certain F_n for some large enough n . Therefore, $y = x - \sum_{i=1}^n \lambda_i e_i$ is also in F_n , and $3\lambda_{n+2} \leq d(\lambda_{n+1} e_{n+1}, F_n) \leq |\lambda_{n+1} e_{n+1} - y| \leq \sum_{i=n+2}^{\infty} \lambda_i \leq \frac{3}{2} \lambda_{n+2}$. This is absurd since $\lambda_n > 0$ for all n , since $e_n \notin F_{n-1}$.

The following is a slight extension of Banach's fixed point theorem in complete spaces.

Suppose that (E, d) is a compact metric space and $f : E \rightarrow E$ is such that $d(f(x), f(y)) < d(x, y)$ for all $x, y \in E$. Show that f has a fixed point.

Proof. If the inequality comes with a factor $d(f(x), f(y)) < Ld(x, y)$ with $L < 1$, this is the classical Banach's fixed point theorem in complete spaces.

In our case, consider the continuous map $x \mapsto d(x, f(x))$ from E to $\mathbb{R}$. This map takes its minimum at some point $\alpha \in E$: consider $f(\alpha)$, then $d(f(\alpha), f(f(\alpha))) < d(\alpha, f(\alpha))$ if $\alpha \neq f(\alpha)$. Therefore α is a fixed point of f .

It is easy to show then that the fixed point is unique.

8.4 Baire category theorem

Show that a normed vector space E with an infinite countable basis cannot be complete.

Proof. Take a basis $\{e_n\}_{n \geq 1}$ and consider $F_n = \text{span}\{e_1, \dots, e_n\}$. Since E is infinite dimensional, F_n is of empty interior, and their union is E , which is of non-trivial empty interior. If E were complete, the Baire category theorem tells us that one of the F_n is of non-empty interior, which is a contradiction.

Let $(E, d), (F, \delta)$ be two metric spaces with E complete. Suppose that a sequence of continuous functions $f_n : E \rightarrow F$ converges pointwise to a function $f : E \rightarrow F$.

1. For all $\epsilon > 0$ and $n \in \mathbb{N}$, define

$$F_{n,\epsilon} = \{x \in E ; \forall p \geq n, \delta(f_p(x), f_n(x)) \leq \epsilon\}.$$

Use Baire to show that Ω_ϵ , the union of the interiors of all $F_{n,\epsilon}$ for $n \in \mathbb{N}$, is dense in E .

2. Show that for all $x_0 \in \Omega_\epsilon$, there is some neighborhood V of x_0 such that for all $x \in V$, $\delta(f(x_0), f(x)) \leq 3\epsilon$.
3. Show that the set of continuity of f is dense.
4. Show that the set of continuity of the derivative $f' : \mathbb{R} \rightarrow \mathbb{R}$ of a derivable function f is dense in $\mathbb{R}$.

Proof. 1. First notice that each $F_{n,\epsilon}$ is closed by continuity of f . Indeed, for each fixed $p \geq n$, the set $\{x \in E ; \delta(f_p(x), f_n(x)) \leq \epsilon\}$ is closed by continuity and $F_{n,\epsilon}$ is their intersection. Also, notice that the union of all $F_{n,\epsilon}$ is E since f_n converges simply.

Now consider the complement of the union of the interiors of $F_{n,\epsilon}$ in E : we denote this by G , which is closed. The intersection of G with $F_{n,\epsilon}$ is closed and of empty interior, and their union is G . By Baire category theorem, G is of empty interior and we are done.

2. By the first question, there is some $x_0 \in \Omega_\epsilon$. For each x_0 , by definition it is in the interior of each $F_{n,\epsilon}$, so that
3. Define $\Omega = \bigcap_{n \in \mathbb{N}} \Omega_{1/n}$.

8.5 More exercises from the web

W2. Nilpotence.

Let $N \in M_n(\mathbb{C})$. Show that N is nilpotent if and only if 0 is adherent to $\{P^{-1}NP, P \in GL_n(\mathbb{C})\}$. Hint: use the upper-triangular representation and the continuity of the characteristic polynomial.

W4. Untitled.

Let $A \in M_n(\mathbb{R})$. Define the function $A^* : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by

$$\forall v \in \mathbb{R}^n, \forall 1 \leq j \leq n, \quad [A^*v]_j = \min\{v_i + A_{ij} ; 1 \leq i \leq n\}.$$

Show that A^* is Lipschitz, then show that for all $\lambda \in (0, 1)$, there exists a unique $v_\lambda \in \mathbb{R}^n$ such that $v_\lambda = A^*(\lambda v_\lambda)$.

W5. Show that $\mathbb{R}^2$ is not a countable union of C^1 curves. Hint: use Baire.

8.6 More exercises from textbooks

T1. Topological monotonicity.

This exercise is taken from the book “Elliptic Partial Differential Equations and Quasiconformal Mappings in the Plane” by Astala, Iwaniec, and Martin.

Show that a map $u : \mathbb{R} \rightarrow \mathbb{R}$ is monotonic if and only if the inverse image of each point $u^{-1}(\{x\})$ is connected.

A mapping $f : X \rightarrow Y$ between two compact metric spaces is called topologically monotone if f is continuous, onto and $f^{-1}(a)$ is connected in X for all $a \in Y$. Prove the following theorem of Whyburn: if f is topologically monotone, then for each closed and connected subset $A \subset Y$, the inverse image $f^{-1}(A) \subset X$ is closed and connected.

Suppose now that f_j is a sequence of topologically monotone mappings converging uniformly to $f : X \rightarrow Y$. Then f is monotone.

T2.